

Sharayu Rasal

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Education

- New York University**, Master of Engineering in Computer Science (GPA: 3.78/4.0) Sep 2024 – May 2026
- **Coursework:** Cloud Computing, Design & Analysis of Algorithms, Software Engineering, Machine Learning, Deep Learning
- AISSMS College of Engineering**, B.E. in Computer Science (Honors in Data Science) (CGPA: 8.15/10.0) Aug 2020 – May 2024
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Web Technology, Operating Systems, Natural Language Processing, Database Management Systems, Artificial Intelligence, Distributed Systems, Business Intelligence

Skills

Languages: Python, C++ , JavaScript, TypeScript, SQL

Frontend: React.js, React Native, Next.js, Redux, HTML5, CSS3, WebSockets

Backend & Frameworks: Django, FastAPI, Flask, Node.js, REST APIs, GraphQL, Kafka, gRPC

Distributed Systems & Cloud: AWS (Lambda, S3, DynamoDB, EC2, EKS, CloudFormation, APIGateway, Amplify, Comprehend, Textract, Bedrock, SQS, CodePipeline), Docker, Kubernetes, Redis, Celery, Hadoop MapReduce, Apache Spark

Databases: PostgreSQL, MongoDB, MySQL, Pinecone

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, Pandas, NumPy

Tools & Infra: Git/GitHub, Linux, CI/CD (Docker-based pipelines, GitHub Actions), kubectl, Prometheus, CloudWatch, OpenSearch

Professional Experience

- Research & Development Intern, Bharat Forge Ltd (KCTI)** Pune, India Jul 2023 – Nov 2023
- Designed and optimized a **real-time computer vision pipeline** using **Python and OpenCV**, reducing end-to-end detection latency by **18%** in resource-constrained unmanned systems.
 - Implemented **custom data transmission protocols** to improve reliability and throughput under strict bandwidth constraints.
 - Collaborated with cross-functional teams, including **DRDO engineers**, to integrate embedded modules and document **API contracts and hardware interfaces** for production deployment.
- Software Engineering Intern, Posit Source Technologies** Pune, India Mar 2023 – June 2023
- Architected and automated **scalable ETL pipelines** in **Python** to process large logistics datasets, ensuring data consistency for downstream systems.
 - Developed and deployed **RESTful APIs** and internal dashboards for real-time analytics, reducing manual reporting effort by **60%**.
 - Improved system performance by refactoring **Redis-based caching** and scheduling logic, reducing delivery lead time by **30%**.
 - Contributed in an **Agile development environment**, participating in **design discussions, code reviews**, and sprint planning to ensure scalability and maintainability.
- Technical Product Intern – Web & Marketing, Oniv Beverages Pvt Ltd** Mumbai, India Dec 2023 – May 2024
- Built and deployed a **React-based customer-facing web application**, improving performance and contributing to a **30% increase in organic traffic**.
 - Implemented **client-side event instrumentation** and integrated it with an **AWS Kinesis ingestion pipeline** to enable real-time analytics and funnel analysis.

Selected Projects

- SmartCache AI (Distributed Content Recommendation System)** 🔄
- Designed and implemented a **distributed content preparation system** using **Django, React, and PostgreSQL** to curate and deliver personalized offline media based on user preferences and commute windows.
 - Built an **event-driven ingestion and processing pipeline** using **Celery and Redis**, supporting scheduled ETL, deduplication via GUID hashing, background downloads, and storage policy enforcement across multiple content sources.
 - Implemented a **real-time multi-agent orchestration workflow** with **AutoGen, WebSockets (Django Channels), and async workers**, enabling live coordination of content discovery, queuing, and download execution with low-latency user feedback.
- LoRA-RoBERTa News Classification App** 🔄
- Fine-tuned a **RoBERTa-based text classifier** using **LoRA**, achieving **94.6% accuracy** with only **0.7% trainable parameters**.
 - Deployed a **high-throughput inference service** using **FastAPI and Docker**, delivering **sub-150 ms** response times through a React-based interface.
 - Reduced **GPU memory usage by ~95%** by applying parameter-efficient fine-tuning and lightweight model serving techniques.
- Research Paper RAG System** 🔄
- Designed a **serverless, multi-tenant document intelligence system** using **AWS Lambda, API Gateway, DynamoDB, and S3**, enabling scalable ingestion and querying of academic papers.
 - Implemented an **asynchronous document processing pipeline** using **SQS**, decoupling upload, extraction, chunking, and indexing to ensure fault tolerance and consistent query behavior.
 - Built a **low-latency retrieval-augmented generation workflow** using **vector embeddings and semantic search**, returning citation-backed responses while enforcing per-user data isolation.

Leadership

- **Global Ambassador, New York University:** Represented NYU at graduate admissions and outreach events, independently guiding prospective students through program structure, coursework, and campus resources.